

2024

Mercedes-Benz Consulting

Creating a Q&A Digital Assistant Using Amazon SageMaker with Mercedes-Benz Consulting

Learn how Mercedes-Benz Consulting in the automotive industry created an AI Q&A system using Amazon SageMaker.

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Millions

of technical documents made searchable

Improved

staff productivity

Sped up

time to market

Achieved good performance

and solution accuracy



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Overview

[Mercedes-Benz Consulting](#), a subsidiary of

giving them better access to company and project information. Whether they are working on a client project or internally, Mercedes-Benz Consulting employees often need to obtain answers quickly from so millions of technical documents from various source systems, and searching these needs to be



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



In 2022, the company decided to build an artificial intelligence (AI)–powered question-and-answer (Q&A) solution using Amazon Web Services (AWS). Mercedes-Benz Consulting worked alongside [AWS Professional Services](#), a global team of experts that assists businesses in realizing desired outcomes when using AWS. Mercedes-Benz Consulting created a digital assistant, as well as other user interface elements and applications, in its Q&A solution so that Mercedes-Benz Consulting employees could access knowledge in a simple, searchable interface. Using AWS, Mercedes-Benz Consulting sped up the time to market for its solution, achieved good performance and accuracy of answers, and improved staff productivity.



Opportunity | Working alongside AWS Professional Services to Design an AI Q&A Solution for Mercedes-Benz Consulting

Mercedes-Benz Consulting is focused on developing data science and AI solutions that are based on cloud technology. The company wanted to create a custom solution for document searching so that Mercedes-Benz Consulting employees could find answers—from how to order a company car to who on an internal development team knows Python—by asking questions in natural language.

With millions of internal documents to search through, getting relevant answers quickly was a standing challenge for employees. Many of these documents were of diverse types of content and data formats and sometimes contained images or videos alongside text. Additionally, if an employee left the company or a colleague with specific knowledge was not available, the only way to find the needed information was within these documents.

To resolve these challenges, Mercedes-Benz Consulting decided to build a Q&A system using machine learning (ML). The company first attempted a solution in 2016, but AI technology had not advanced enough to achieve its goals. In 2021, the company created a proof of concept using AWS to make its solution flexible and scalable, and in 2022, the company engaged AWS Professional Services to work together on its solution for 6 months. “For the developer team, AWS Professional Services played a crucial role in professionalizing our cloud, data, and ML infrastructure on AWS,” says Gavneet Singh Chadha, management consultant at Mercedes-Benz Consulting. AWS Professional Services also played a role in expanding skill development for the Mercedes-Benz Consulting +

Solution | Improving Staff Productivity by Making Millions of Documents Searchable with AI

Mercedes-Benz Consulting needed to architect a strong foundation for its data sources, and it took a serverless-first approach. When employees ask a question of the AI solution, the process starts with data going through the extract, transform, load process. To automate the migration of data, Mercedes-Benz Consulting uses [AWS Lambda](#), a serverless, event-driven compute service. This data is then processed and cleaned using [AWS Glue](#), a serverless data integration service that makes it simple to discover, prepare, move, and integrate data from multiple sources.

Another important component of the solution is the processing of natural language. Here, text needs to be converted to numbers using embeddings. For its ML model development, monitoring, and deployment, Mercedes-Benz Consulting uses [Amazon SageMaker](#), a service to build, train, and deploy ML models. The company can also use Amazon SageMaker to integrate open-source ML models from Hugging Face while keeping control of its data. These data embeddings are then accessed by [Amazon OpenSearch Service](#), which is used to securely unlock near-real-time search, monitoring, and analysis of business and operational data. The embeddings then go through a vector search, and the AI solution provides the answer to the initial question.

The speed and accuracy of this solution helps improve staff productivity at Mercedes-Benz Consulting. The company's goal is for the top three returned results of any query to contain all the relevant information, with the first result providing the best answer to the question. One metric for measuring accuracy is the percentage of questions with positive user feedback from qualified users. The speed and accuracy of answers saves time and effort for employees, especially new hires, when searching for information. "People working in the company have gained knowledge over years that can be difficult to quantify," says Dr. Chadha. "Having a system provide this information in a simple, near-real-time way is massive for the company's growth and productivity."

By building this AI product, the company helps drive innovation at Mercedes-Benz Consulting. Developers can work more efficiently using the AI assistant, and searching for information by asking questions in natural language eases workflow friction across the company. Additionally, using AWS for this solution reduced its time to market.

"Creating this solution would have taken much longer if we did not have the support of AWS Professional Services," says Dr. Chadha.



“Using AWS, we created a cutting-edge system that allows our company to bring our ways of using internal knowledge to a completely new level,” says Dr. Marcel Graus, head of data and tech at Mercedes-Benz Consulting.

Outcome | Designing AI Solutions for Additional Use Cases

Mercedes-Benz Consulting is in the process of developing innovative solutions for multiple internal clients using mature AI technology that is flexible, scalable, and secure. Although the new application is already in use within Mercedes-Benz Consulting, the company wants to expand its solution for use in workshops and on showroom floors.

“With this AWS engagement and everything we’ve developed in the past year, we have moved our focus from doing projects to creating products,” says Dr. Chadha. “We have learned to build in a flexible, scalable manner so that we can incorporate user requirements and develop our system further.”

About Mercedes-Benz Consulting

Mercedes-Benz Consulting performs consulting and technology training and implementation for the Mercedes-Benz Group AG. It implements AI and data science solutions for clients in the cloud and offers consulting to conceptualize solutions for clients.

AWS Services Used

Amazon SageMaker

Build, train, and deploy machine learning (ML) models for any use case with fully managed infrastructure, tools, and workflows.

[Learn more »](#)

AWS Glue

Discover, prepare, and integrate all your data at any scale.

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AWS Professional Services

AWS Professional Services' offerings use a unique methodology based on Amazon's internal best practices to help you complete projects faster and more reliably, while accounting for evolving expectations and dynamic team structures along the way.

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Amazon OpenSearch Service

Securely unlock real-time search, monitoring, and analysis of business and operational data.

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EUROPE, MIDDLE EAST, & AFRICA



Continental on AWS

From one of the most recognized tire brands in the world to a connected mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to the cloud and provides a virtual

2024



ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

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